

Student Placement Salary Analysis

Group 3: Nafisa Nawar Aurny, Uzayr Chaudhry, David Mwaria, Ruth Picazo

Data Description.....	2
Introduction.....	2
Data Cleaning.....	2
Table 1. Cleaning summary.....	2
Table 2. Variables used in the salary analysis.....	3
Exploratory Analysis.....	3
Data Analysis.....	7
Variable Selection.....	7
Table 3. Full vs Selected Model.....	9
Residual Analysis.....	10
Table 4a. OLS vs Tobit Model.....	11
Table 4b. Selected vs. Interaction Model.....	13
Transformations.....	14
Table 5. Tested transformations on interaction OLS.....	14
Table 6. Full vs Stepwise vs Interaction vs Final Model.....	18
Conclusion.....	18
Reflective Process.....	19
Appendix.....	20
References.....	20
Member Roles.....	20
R Scripts and Reproducibility Materials.....	21
Key Output Materials.....	21
Final Model Formula.....	22
Selected R Code Excerpts.....	22
Code A1. Data preparation and placed-students-only modeling data set.....	22
Code A2. Variable selection and VIF check.....	23
Code A3. Interaction check, residual diagnostics, and transformation check.....	24
Code A4. Final model used for interpretation.....	24
Code A5. Predicted salary tables for interpreting interactions.....	25
Code A6. Final-model residual diagnostics figure.....	25

Data Description

Introduction

The goal of this project is to determine which student-related factors best explain salary among students who received job placements. We used the Global Student Placement & Salary Dataset from Kaggle, which includes information about students' academic performance, skills, internship experience, university background, placement industry, placement status, and salary. The main research question is: Which student-related factors best explain salary among students who received job placements?

The original dataset contains 10,000 observations and 13 variables. Since students who were not placed had a salary recorded as zero, we restricted the analysis to only students who were placed. This avoids mixing placement prediction with salary prediction. After filtering, the final modeling dataset contains 6,153 placed students. The response variable is salary, and the predictors include CGPA, backlogs, internship count, aptitude score, communication score, internship quality score, college tier, country, university ranking band, specialization, and industry.

Data Cleaning

The data set was cleaned in R prior to analysis. The original CSV file was first read into R. We then filtered the data to only include observations where `placement_status = "Placed"`. Following this, `placement_status` was removed as no other student had a different placement status, and thus did not provide any additional information to explain the differences in salary.

The cleaned dataset was checked for missing values and duplicate observations. No missing values or duplicates were found. Finally, categorical variables such as `college_tier`, `country`, `university_ranking_band`, `specialization`, and `industry` were converted into factors so they could be used correctly in the regression models.

Table 1. Cleaning summary	
Step	Result
Original Dataset	10,000 observations, 13 variables
Filter Applied	<code>placement_status = "Placed"</code>

Table 1. Cleaning summary	
Final modeling dataset	6,153 placed students
Removed variable	placement_status
Missing values	None found
Duplicate observations	None found
Categorical variables	Converted to factors

Table 2. Variables used in the salary analysis		
Variable	Role	Type
salary	Response	Continuous
cgpa	Predictor	Continuous
backlogs	Predictor	Discrete numeric / count
internship_count	Predictor	Discrete numeric / count
aptitude_score	Predictor	Continuous
communication_score	Predictor	Continuous
internship_quality_score	Predictor	Continuous
college_tier	Predictor	Categorical
country	Predictor	Categorical
university_ranking_band	Predictor	Categorical
specialization	Predictor	Categorical
industry	Predictor	Categorical

Exploratory Analysis

Before fitting regression models, exploratory data analysis was used to understand the data set. The salary histogram revealed that there was a tendency for salaries to be grouped around the high end, with a definite ceiling of 120,000. This cap is crucial because it can impact

the remaining trend in the regression model. The boxplots also showed that salary differs strongly by country and college tier, suggesting that these categorical variables are important predictors.

Scatterplots were used to examine salary against numeric predictors such as CGPA, internship count, aptitude score, communication score, and internship quality score. Overall, most numeric predictors showed weak individual linear relationships with salary. However, when salary was viewed by country, clear group differences appeared, suggesting that categorical background variables explain more salary variation than most numeric predictors alone.

Figure 1. Distribution of salary for placed students.

The histogram shows that many observations are near the upper salary range, with a visible cap at 120,000.

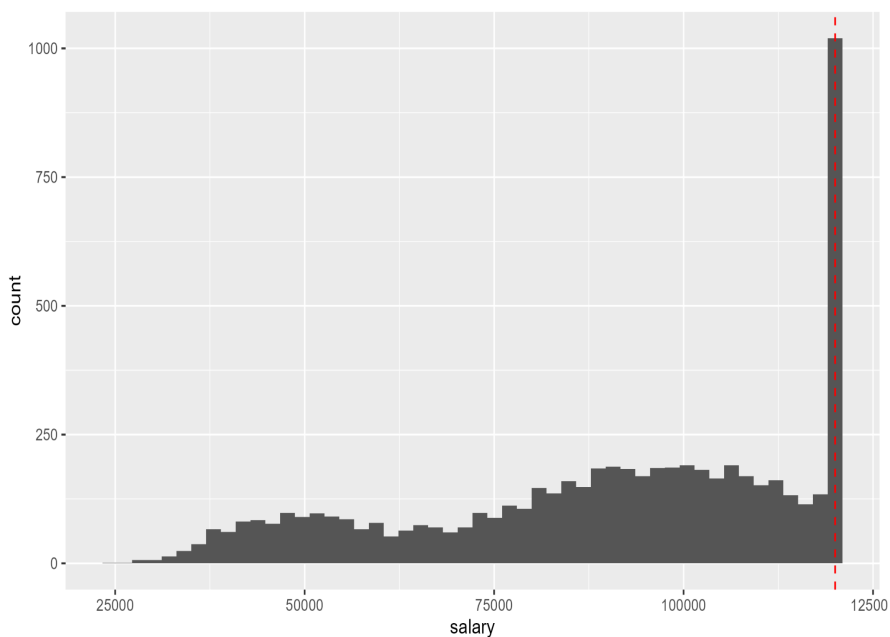


Figure 2. Salary by country.

The boxplot shows clear differences in salary distributions across countries.

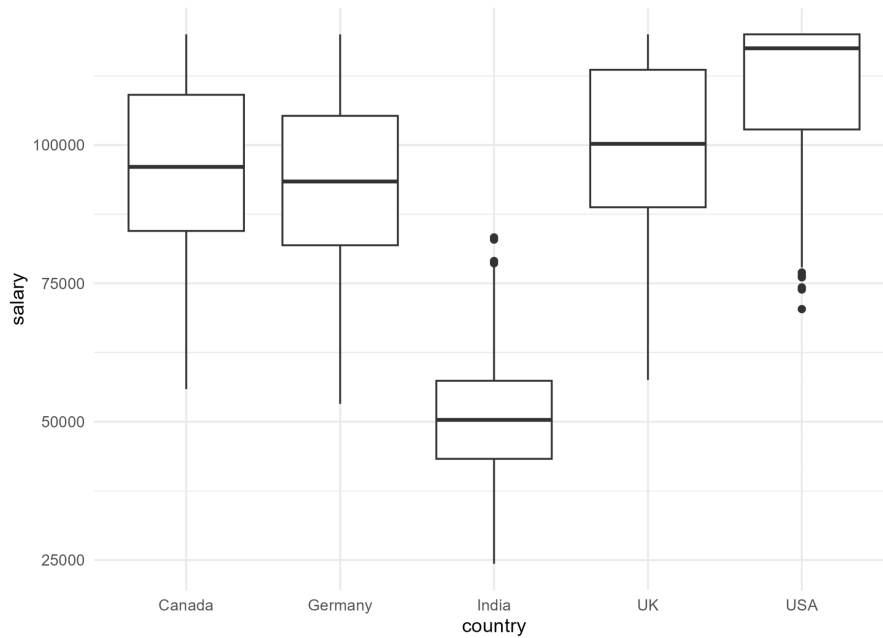


Figure 3. Salary by college tier.

The boxplot shows that salary tends to decrease from Tier 1 to Tier 3 colleges.

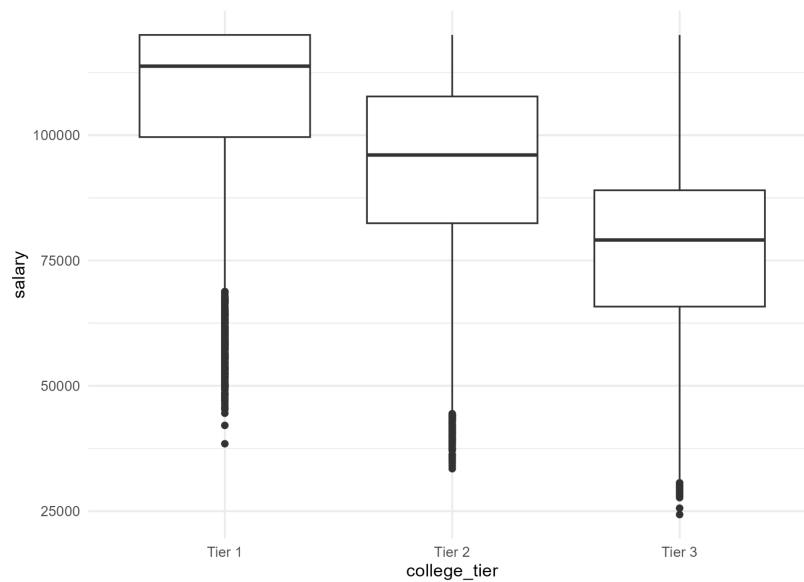


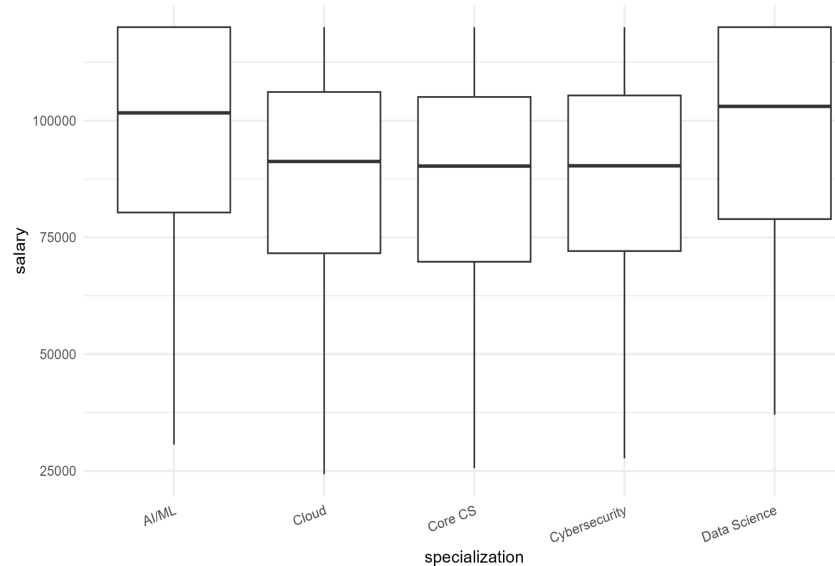
Figure 4. CGPA vs. salary by country.

The scatterplot shows that salary differences are more clearly separated by country than by CGPA alone.



Figure 5. Salary by specialization.

The boxplot shows that salary differs somewhat across specializations, although the differences are not as strong as the country and college-tier patterns.



Overall, the exploratory analysis suggests that salary is more strongly separated by categorical variables, especially country and college tier, than by most numeric predictors alone. The visible salary cap of 120,000 is also important because it may affect the regression residuals. These findings motivate the use of multiple regression models that can evaluate numeric and categorical predictors together.

Data Analysis

Variable Selection

We began with our “full model,” which included the 11 predictors. Although the full model yielded a high adjusted R^2 of 0.9466, the summary output (Figure 6) indicated that several variables (backlogs, internship count, internship quality, and communication score) were not statistically significant. A check of Variance Inflation Factors (VIF) displayed that all values were below 5 (Figure 7), indicating no severe multicollinearity. Therefore, variables were not removed due to multicollinearity.

Figure 6. Full Model Summary

```
Call:
lm(formula = salary ~ cgpa + backlogs + college_tier + country +
    university_ranking_band + specialization + industry + internship_count +
    internship_quality_score + aptitude_score + communication_score,
    data = df_placed)

Residuals:
    Min       1Q   Median       3Q      Max
-19157.9  -3876.4   152.7   4089.1  17379.0

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      81523.494    978.560   83.310  <2e-16 ***
cgpa              4340.890    100.664   43.123  <2e-16 ***
backlogs          17.006     78.669    0.216    0.829
college_tierTier 2 -12004.327    176.789  -67.902  <2e-16 ***
college_tierTier 3 -27527.263    189.474 -145.283  <2e-16 ***
countryGermany    -3584.236    232.793  -15.397  <2e-16 ***
countryIndia     -45845.539    231.898 -197.697  <2e-16 ***
countryUK         3396.546    235.374   14.430  <2e-16 ***
countryUSA       14069.448    231.019   60.902  <2e-16 ***
university_ranking_band300+ -2222.131    167.841  -13.239  <2e-16 ***
university_ranking_bandTop 100 2689.660    191.578   14.039  <2e-16 ***
specializationCloud -9778.895    231.497  -42.242  <2e-16 ***
specializationCore CS -10149.481    234.114  -43.353  <2e-16 ***
specializationCybersecurity -9908.113    232.245  -42.662  <2e-16 ***
specializationData Science -263.809    232.175   -1.136    0.256
industryFinance   3676.700    249.692   14.725  <2e-16 ***
industryHealthcare 287.737    248.614    1.157    0.247
industryManufacturing 358.263    258.169    1.388    0.165
industryOther      90.386    259.061    0.349    0.727
industryTech      6742.662    246.701   27.331  <2e-16 ***
internship_count    6.798     59.251    0.115    0.909
internship_quality_score 32.267     50.061    0.645    0.519
aptitude_score      7.662     5.072    1.511    0.131
communication_score  3.829     4.998    0.766    0.444
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5752 on 6129 degrees of freedom
Multiple R-squared:  0.9468, Adjusted R-squared:  0.9466
F-statistic: 4746 on 23 and 6129 DF, p-value: < 2.2e-16
```

Figure 7. VIF

	GVIF
cgpa	1.059251
backlogs	1.052334
college_tier	1.007133
country	1.013564
university_ranking_band	1.013531
specialization	1.013047
industry	1.024403
internship_count	1.011771
internship_quality_score	1.021103
aptitude_score	1.004961
communication_score	1.005536

Using a backwards elimination approach with significance level $\alpha = 0.05$, we removed the insignificant variables: backlogs, internship count, internship quality, and communication score. This left us with the selected model: $\text{salary} \sim \text{cgpa} + \text{aptitude_score} + \text{college_tier} + \text{country} + \text{university_ranking_band} + \text{specialization} + \text{industry}$.

Figure 8. Categorical Predictors

predictor	r_squared	adj_r_sq	f_statistic	df_model	f_p_value
college_tier	0.1967	0.1965	753.111	2	0
country	0.6978	0.6976	3549.087	4	0
university_ranking_band	0.0038	0.0035	11.813	2	0
specialization	0.0304	0.0298	48.193	4	0
industry	0.0074	0.0066	9.145	5	0

Figure 9. Numerical Predictors

predictor	coefficient	std_error	t_statistic	p_value	r_squared	adj_r_sq	f_statistic	f_p_value
cgpa	3383.703	421.252	8.032	0	0.0104	0.0102	64.521	0
backlogs	-966.14	331.791	-2.912	0.0036	0.0014	0.0012	8.479	0.0036
internship_count	47.667	255.028	0.187	0.8517	0	-2e-4	0.035	0.8517
aptitude_score	-38.634	21.899	-1.764	0.0778	5e-4	3e-4	3.112	0.0778
communication_score	-10.265	21.578	-0.476	0.6343	0	-1e-4	0.226	0.6343
internship_quality_score	40.932	214.484	0.191	0.8487	0	-2e-4	0.036	0.8487

The next step was to do a model comparison (Table 3) to verify that the selected model was preferred over the full model. The selected model performs essentially the same as the full model, but with fewer variables, a slightly lower AIC, and nearly identical adjusted R².

Full model: salary ~ cgpa + backlogs + college_tier + country + university_ranking_band + internship_count + aptitude_score + communication_score + specialization + industry + internship_quality_score

Selected model: salary ~ cgpa + aptitude_score + college_tier + country + university_ranking_band + specialization + industry

Table 3. Full vs Selected Model			
Model	AIC	Adj. R ²	Terms
Full	124024.16	0.946642	11
Selected	124017.23	0.946667	7

Figure 10. ANOVA Table

ANOVA Table of the full model and the selected model

Analysis of Variance Table

Model 1: salary ~ cgpa + backlogs + college_tier + country + university_ranking_band + internship_count + aptitude_score + communication_score + specialization + industry + internship_quality_score

Model 2: salary ~ cgpa + aptitude_score + college_tier + country + university_ranking_band + specialization + industry

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	6129	2.0278e+11				
2	6133	2.0282e+11	-4	-35238488	0.2663	0.8998

Residual Analysis

Since it was noted that the salary cap influenced our results, the residuals on the fitted salaries under \$120,000 aren't constant around 0, as there is a noticeable upward-bowl shape (Figure 11). We decided to fit a Tobit model as a sensitivity check for the salary ceiling. A tobit model separates the data into values below the ceiling and those at the ceiling and adjusts the coefficients on the predictors accordingly. This model produced coefficients of greater magnitude. Compared to OLS, the magnitude of the coefficients increases because the linear model is not being diluted by the ceiling (Table 4a). However, the Tobit model was not used to select a new final set of predictors. Instead, it served as a sanity check showing that the salary cap was affecting coefficient estimates and residual behavior. Although the residuals of the tobit model look marginally better, there is still a curvature (Figure 12).

Figure 11. Residual Diagnostic of Selected Model

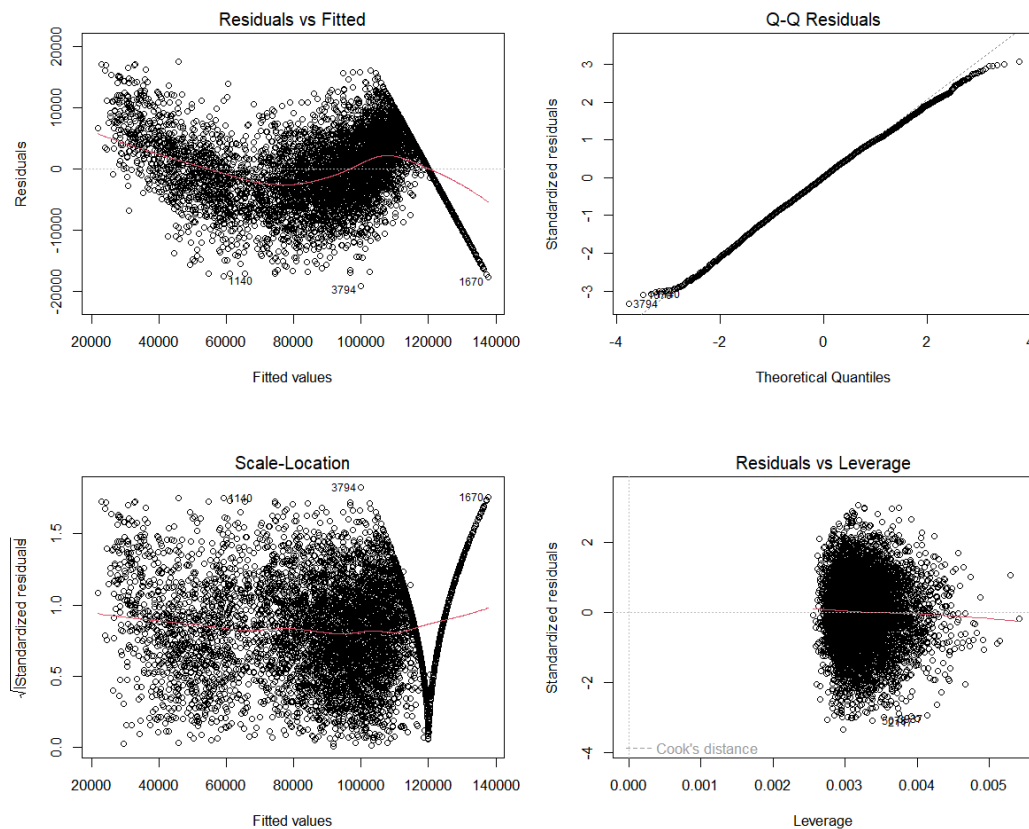


Figure 12. Residual vs Fitted (Tobit Model)

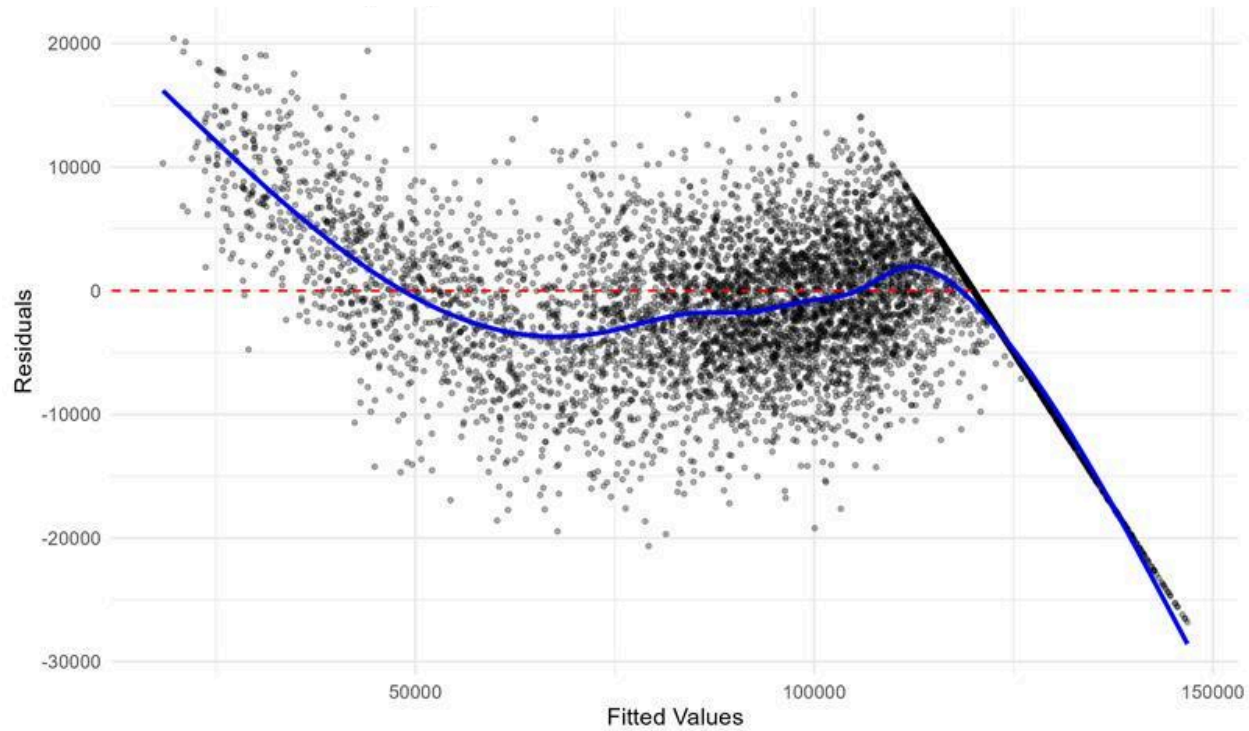


Table 4a. OLS vs Tobit Model		
	OLS	Tobit
(Intercept)	81988.817074	81231.215913
cgpa	4339.221062	4940.320317
aptitude_score	7.626207	7.998635
college_tierTier 2	-12004.744219	-14869.407823
college_tierTier 3	-27526.380348	-30971.879642

We also found that the college tier and country affect salary (Figure 13). For example, a high-tier university in the USA has a different impact on outcomes than a high-tier university in

India. We then compared our selected (base) model and a model with the interaction of college tier by country. This yielded a P value < 0.005 (Figure 14) and a higher adjusted R^2 for the interaction model (Table 4b), revealing that the interaction model is better than the base model.

Figure 13. Mean Salary by College Tier and Country

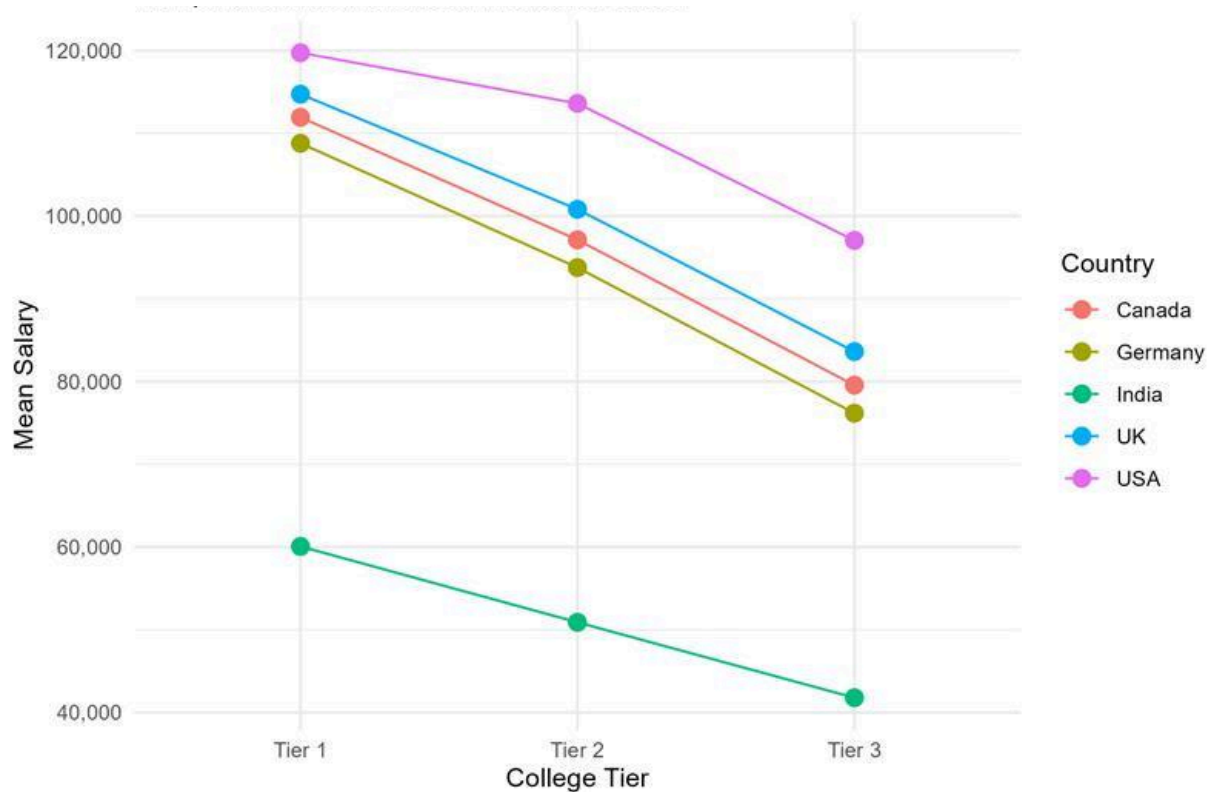


Figure 14. ANOVA Table

ANOVA Table of the base selected model and the interaction model

Analysis of Variance Table

Model 1: salary ~ cgpa + aptitude_score + college_tier + country + university_ranking_band + specialization + industry

Model 2: salary ~ cgpa + aptitude_score + college_tier * country + university_ranking_band + specialization + industry

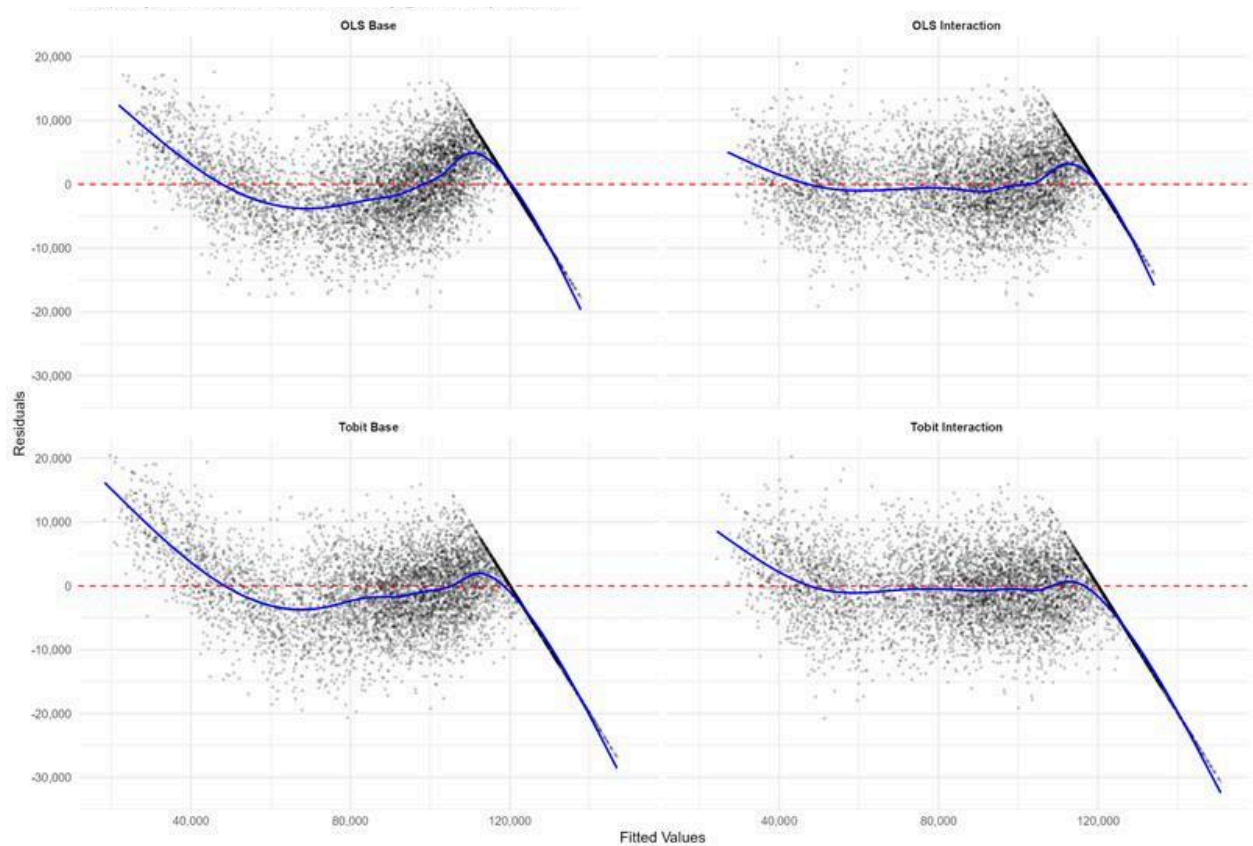
	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	6133	2.0282e+11				
2	6125	1.6411e+11	8	3.8703e+10	180.56	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 4b. Selected vs. Interaction Model			
Model	AIC	Adj. R ²	Terms
Base	124017.23	0.946667	7
Interaction	122730.39	0.956978	7

Accounting for this interaction produced a residual pattern that was more centered and less curved than the selected additive model. By comparing the four models (Figure 15), OLS and Tobit with and without interactions, the curvature is reduced after adding the tier-by-country interaction.

Figure 15. Residual vs Fitted: OLS vs Tobit



Transformations

We performed a Box-Cox on OLS with interactions (Figure 16), and our results yielded a Box-Cox $\lambda = 0.9$, suggesting little or no transformation is needed. To solidify our conclusion, we tested the transformations on OLS with the interaction of the square root and log transformations on salary (Table 5). Please note that the arcsine transformation was not tested because salary is not a 0–1 proportion variable.

Figure 16. Box-Cox Profile for Interaction OLS

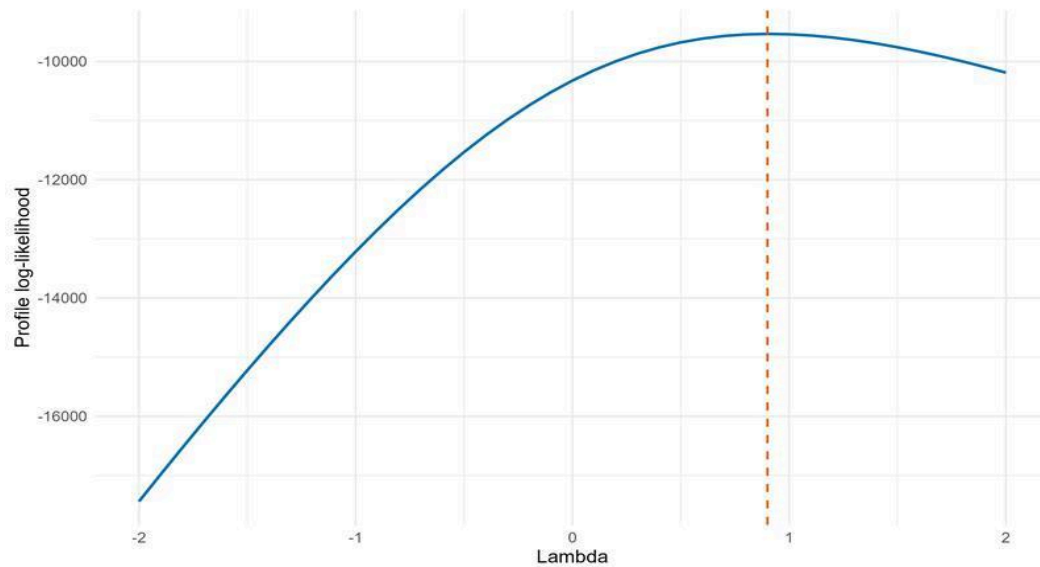


Table 5. Tested transformations on interaction OLS

Model	Response Variable	Residual SD	Adjusted R ²	Note
raw_salary_interaction_ols	salary	5,176.291	0.956788	Main model: easiest to interpret in salary dollars
sqrt_salary_interaction_ols	sqrt(salary)	9.027792	0.958888	Possible variance-stabilizing check; less common for salary

Table 5. Tested transformations on interaction OLS				
log_salary_interaction_ols	log(salary)	0.068413	0.956231	Common salary transformation: check plots before using

To check whether a response transformation could reduce the salary-cap pattern, we tested $\log(\text{salary})$ as a response transform. Residuals are a bit flatter in the middle, but the high-end cap pattern remains (Figure 17), and the QQ tails were less than ideal compared to the raw salary interaction OLS model (Figure 18).

The raw salary is best for communication and business interpretation because the coefficients stay in dollar terms. Because the transformed models use different response scales, their residual standard deviations and adjusted R^2 values should not be interpreted as directly comparable to the raw salary model. Instead, the transformations were used as diagnostic checks. The Box-Cox result gave $\lambda \approx 0.9$, which is close to 1, and the log-salary residual plot still showed the high-end cap pattern. As such, the raw salary will be used as it is simpler and easier to interpret.

Figure 17. $\log(\text{Salary})$ Interaction OLS: Residuals vs. Fitted salary

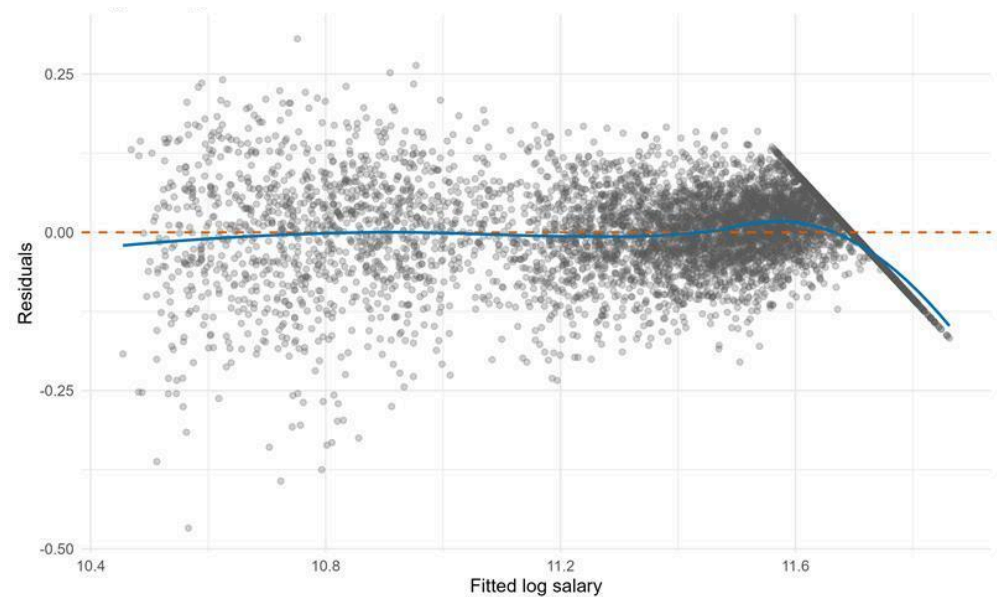
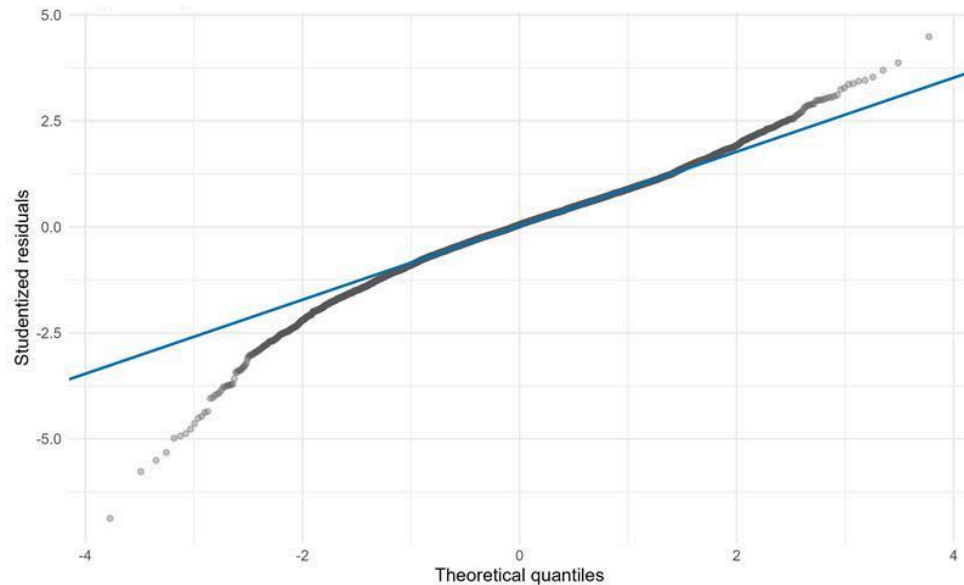


Figure 18. Log Salary Interaction OLS: QQ Plot



We then tested additional interactions to reduce the remaining residual pattern and found that adding the interaction between country and specialization was the most justified. The mean salary plot suggested that specialization effects differed by country (Figure 19). The final residual diagnostics improved relative to earlier models (Figure 20), and the ANOVA comparison for adding the new interaction was statistically significant (Figure 21). Based on these results and the model comparison in Table 6, we selected this as the final model.

Figure 19. Mean Salary by Country and Specialization

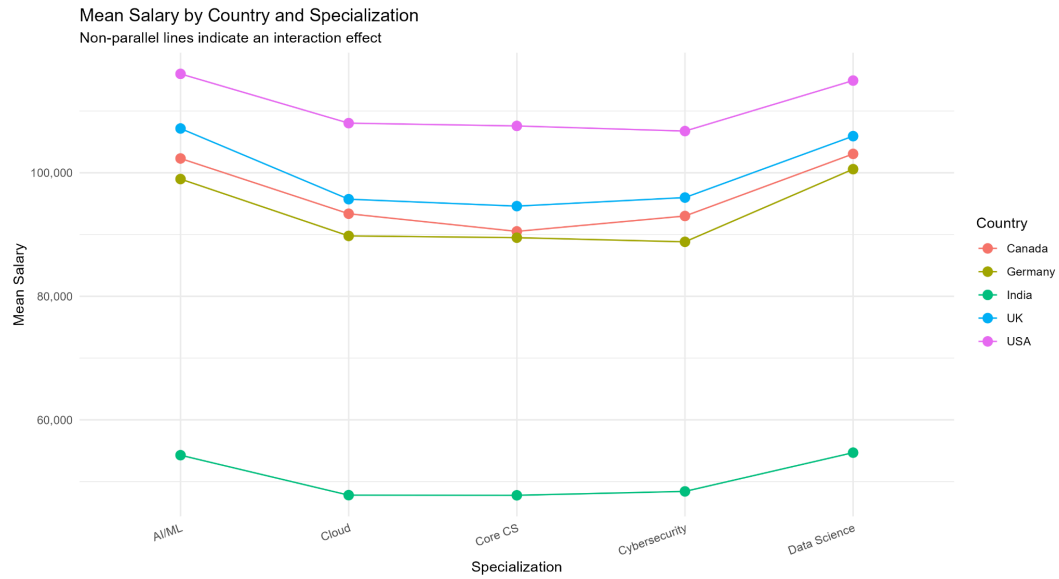


Figure 20. Residual Diagnostics of Final Model

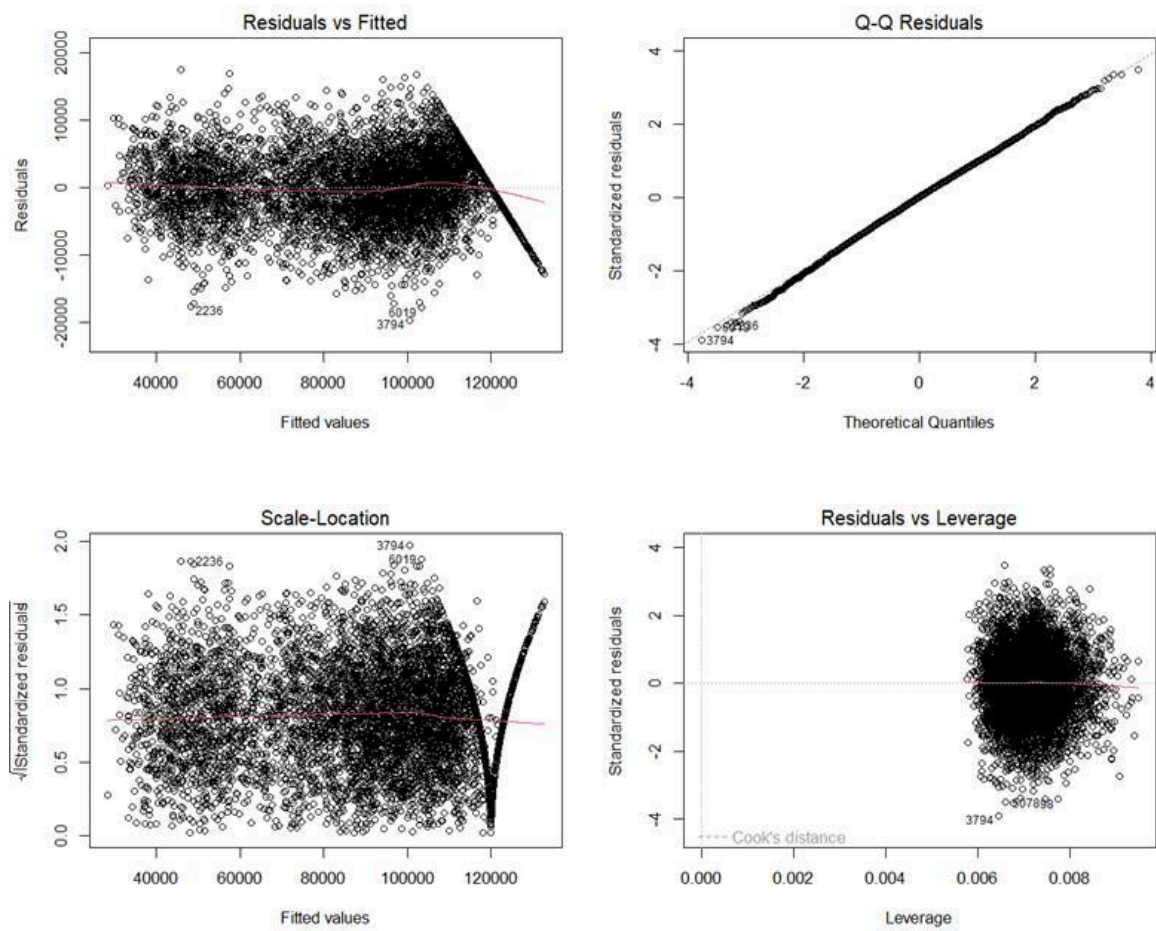


Figure 21. ANOVA Table

ANOVA Table of the 1 interaction model and the 2 interaction model

Analysis of Variance Table

Model 1: salary ~ cgpa + aptitude_score + college_tier * country + university_ranking_band + specialization + industry

Model 2: salary ~ cgpa + aptitude_score + college_tier * country + country * specialization + university_ranking_band + industry

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	6125	1.6411e+11				
2	6109	1.5596e+11	16	8158132621	19.973	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 6. Full vs Stepwise vs Interaction vs Final Model

Model	Predictors Included	AIC	Adj R ²	Notes
Full Model	All predictors	124024.16	0.946642	Baseline comparison
Stepwise Model	cgpa + aptitude_score + college_tier + country + university_ranking_band + specialization + industry	124017.23	0.946667	Stepwise-selected model
Tier * Country Interaction Model	cgpa + aptitude_score + college_tier * country + university_ranking_band + specialization + industry	122730.39	0.956978	First interaction model
Final Model	cgpa + aptitude_score + college_tier * country + country * specialization + university_ranking_band + industry	122448.65	0.958829	Selected after diagnostics

Conclusion

This analysis focused on explaining salary among students who received job placements. After filtering the original data to placed students only, the final modeling data set contained 6,153 observations. This modeling decision was important because unplaced students had salary recorded as zero; including them would have mixed two different questions: whether a student was placed and how salary differs among students who were placed.

The exploratory analysis showed that salary varied much more clearly across categorical factors than across most numeric predictors alone. Country and college tier showed the strongest visible separation in salary, while numeric predictors such as internship count, communication score, and internship quality score had weak standalone linear relationships with salary. The salary distribution also showed a clear ceiling at 120,000, which remained an important limitation throughout the modeling process.

The final selected OLS model was $\text{salary} \sim \text{cgpa} + \text{aptitude_score} + \text{college_tier} * \text{country} + \text{country} * \text{specialization} + \text{university_ranking_band} + \text{industry}$. This model was chosen after comparing the full model, the selected additive model, the tier-by-country interaction model, and the final model with the additional country-by-specialization interaction. The final model improved the fit compared with the earlier models, with $R\text{-squared} = 0.9591$, adjusted $R\text{-squared} = 0.9588$, residual standard error = 5052.60, and $AIC = 122448.65$. These results indicate that the final model explains a large share of the variation in salary among placed students.

The final model suggests that salary in this data set is most strongly associated with country, college tier, specialization, and how those categorical factors interact. The college tier effect differs by country, and specialization effects also differ by country. CGPA remained a meaningful positive numeric predictor: holding the other variables fixed, a one-point increase in CGPA was associated with an estimated salary increase of about \$4,317. Aptitude score, however, was not statistically significant in the final model after the other variables were included.

The residual analysis supports the usefulness of the interaction terms but also shows that the model is not perfect. Adding interactions made the residual patterns flatter, but the salary cap at 120,000 still produced a visible high-end ceiling effect. Transformation checks did not provide strong evidence to move away from the original salary scale, so salary was kept in dollars for interpretability. Overall, the final model provides a strong and interpretable explanation of placed-student salary patterns, but the salary cap and the absence of employer, city, and cost-of-living variables limit how far the results can be generalized.

Reflective Process

This project provided our team with valuable practice applying several data analysis strategies discussed over the course of the semester. We gained a deeper understanding of conceptual material by working through the process of data cleaning, exploratory data analysis,

predictor selection, and model fitting. While our analysis produced pleasing results, it came with some limitations, namely the salary cap of \$120,000. This ceiling significantly impacted our final model, especially considering the fact that about 9.6% of all observations had a salary at this ceiling. We explored the Tobit model primarily as a sensitivity check for the salary ceiling, and it suggested that the cap affected coefficient estimates. Without this ceiling, we would likely have been able to give a more robust explanation of how the predictors affect salary, and we also may have had several high-salary outliers to account for.

Overall, our progress was relatively smooth because of our initial decision to use a synthetic dataset from Kaggle. The data did not require heavy cleaning, and was friendly to regression techniques with over 10,000 observations, consistent, continuous predictors, and meaningful categorical variables. Real-world data would be messier, likely involving many missing values and perhaps entry errors. Variable selection would be more complicated as well. If we were to build on this project in the future, moving on to a real-world data set would further test our ability to handle ambiguity. We are interested in seeing if the same set of predictors would retain the same power when applied to the real world.

Additionally, we can expand on our work by studying the aspects of education and employment not touched on by our dataset. Our data focused only on the tech field, and excluded any student that did not find job placement directly after graduation. Of course, in reality, there are many other potential options and outcomes. Our results would be more useful to students if they included, for example, analysis of the relationship between the predictors and duration of unemployment/underemployment. We could even consider additional paths, such as graduate school and bootcamps, giving today's students a fuller picture of what their options are.

Appendix

References

- Fox, J., Weisberg, S., Price, B., et al. (2024). *car: Companion to Applied Regression* (R package). CRAN. <https://CRAN.R-project.org/package=car>
- Kaggle. (n.d.). Global Student Placement & Salary Dataset [Data set]. Kaggle. Retrieved May 11, 2026, from <https://www.kaggle.com/datasets/rakesh630/global-student-placement-2025-dataset>
- Kleiber, C., & Zeileis, A. (2024). *AER: Applied Econometrics with R* (R package). CRAN. <https://CRAN.R-project.org/package=AER>
- R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ripley, B., Venables, B., Bates, D. M., Hornik, K., Gebhardt, A., Firth, D., & Ripley, M. B. (2025). *MASS: Support functions and datasets for Venables and Ripley's MASS* (R package). CRAN. <https://CRAN.R-project.org/package=MASS>

- Robinson, D., Hayes, A., Couch, S., & Hvitfeldt, E. (2025). broom: Convert statistical objects into tidy tibbles (R package version 1.0.11). <https://broom.tidymodels.org/>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., & Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>

Member Roles

Member	Contribution
Uzayr	Cleaning, EDA, SLR, AIC, VIF, Box-Cox, final model
David	Residual analysis, Tobit model, interaction terms
Ruth	Data Cleaning, presentation/report writing
Nafisa	Presentation/report writing

R Scripts and Reproducibility Materials

The submitted repository contains the data, R scripts, generated tables, and generated figures used in the analysis. Tables are written to output/tables and figures are written to output/figures. The following list identifies the scripts that should be included with the submitted materials.

- 01_clean_data.R - reads the raw Kaggle data, validates the expected columns, converts data types, filters to placed students, and creates data/global_placement_clean.csv and data/placed_salary_data.csv.
- 02_eda_tables.R - creates numeric summary tables, categorical counts, and the correlation matrix.
- 03_eda_plots.R - creates EDA figures for salary distribution, numeric predictors, and categorical predictors.
- simple_regression_screening.R - fits one-predictor screening models and writes simple_regression_summary.csv.
- model_selection.R - fits the full model, runs stepwise AIC, compares the full and selected models, and writes VIF output.
- model_diagnostics.R - compares the selected base model to the tier-by-country interaction model and writes residual, influence, transformation, and Box-Cox outputs.

- 04_subgroup_residual_checks.R - checks residual patterns by country, country by tier, and country by specialization; also creates severe-overprediction summaries.
- interaction_bar_labeled.R - creates presentation-ready predicted salary figures for the tier-by-country and specialization-by-country model interpretation.
- 07_final_model_coefficients.R - fits the final OLS model, writes final model fit and coefficient tables, creates predicted salary tables, and saves the final model diagnostic plot.
- residual_analysis.R and residual_investigation.R - exploratory scripts used for residual diagnostics, Tobit sensitivity checks, and salary ceiling investigation.
- cgpa_with_country.R and cgpa_country_rank.R - exploratory CGPA visualizations by country and ranking band.

Key Output Materials

Output File	Purpose
slr_numeric_summary.csv	Table of significance of numeric predictors
slr_categorical_summary.csv	Table of significance of categorical predictors
salary_ceiling_check.png	Histogram revealing salary cap at \$120,000
cgpa_vs_salary_by_country.png	Scatterplot visualizing categorical influence
full_model_summary.csv	MLR with all predictors, summary
vif_selected_model.csv	VIF selected predictors
model_comparison.csv	Full model vs selected model comparison
boxcox_summary.csv	Box-Cox summary; no transformation
base_vs_interaction_residuals.png	One interaction term improves residual behavior
resid_all_models.png	Tobit residual comparison: prefer simpler model
predicted_salary_tier_country_gaps.png	Visualize effects of categorical interaction
final_model_residual_diagnostics.png	Final model residuals confirm effective fit

Final Model Formula

The final model used for interpretation was:

$$\text{salary} \sim \text{cgpa} + \text{aptitude_score} + \text{college_tier} * \text{country} + \text{country} * \text{specialization} + \text{university_ranking_band} + \text{industry}$$

This model estimates 44 coefficients total: one intercept plus 43 predictor coefficients. The count is produced by dummy-variable expansion of categorical predictors and interaction terms, not by 44 raw input variables.

Selected R Code Excerpts

The complete R files are included in the project repository. The excerpts below show the main reproducible steps used for the analysis: data preparation, model selection, interaction diagnostics, final model fitting, interpretation outputs, and final residual checks.

Note: The following codes are excerpts. The full script is submitted in the repository; line wrapping below is for report readability.

Code A1. Data preparation and placed-students-only modeling data set

```
library(tidyverse)

raw_df <- read_csv("data/global_placement.csv", show_col_types = FALSE)

clean_df <- raw_df %>%
  mutate(
    across(where(is.character), str_squish),
    cgpa = as.numeric(cgpa),
    backlogs = as.integer(backlogs),
    internship_count = as.integer(internship_count),
    aptitude_score = as.numeric(aptitude_score),
    communication_score = as.numeric(communication_score),
    internship_quality_score = as.numeric(internship_quality_score),
    salary = as.numeric(salary),
    college_tier = factor(college_tier, levels = c("Tier 1", "Tier 2", "Tier 3")),
    country = factor(country, levels = c("Canada", "Germany", "India", "UK", "USA")),
    university_ranking_band = factor(university_ranking_band,
                                     levels = c("Top 100", "100-300", "300+")),
    specialization = factor(specialization,
                           levels = c("AI/ML", "Cloud", "Core CS", "Cybersecurity", "Data Science")),
    industry = factor(industry,
                     levels = c("Consulting", "Finance", "Healthcare", "Manufacturing", "Other", "Tech")),
    placement_status = factor(placement_status, levels = c("Placed", "Not Placed"))
  )

quality_summary <- tibble(
  metric = c("n_rows_raw", "n_cols_raw", "total_missing_values", "duplicate_rows",
            "placed_rows", "not_placed_rows"),
  value = c(nrow(clean_df), ncol(clean_df), sum(is.na(clean_df)), sum(duplicated(clean_df)),
            sum(clean_df$placement_status == "Placed"),
            sum(clean_df$placement_status == "Not Placed"))
)

analysis_df <- clean_df %>%
  filter(placement_status == "Placed") %>%
  select(-placement_status)

write_csv(clean_df, "data/global_placement_clean.csv")
write_csv(analysis_df, "data/placed_salary_data.csv")
write_csv(quality_summary, "output/tables/data_quality_summary.csv")
```

Code A2. Variable selection and VIF check

```
library(dplyr)
library(broom)
library(MASS)
library(car)

placed_data <- read.csv("data/placed_salary_data.csv", stringsAsFactors = FALSE) %>%
  mutate(
    college_tier = as.factor(college_tier),
    country = as.factor(country),
    university_ranking_band = as.factor(university_ranking_band),
    specialization = as.factor(specialization),
    industry = as.factor(industry)
  )

full_model <- lm(
  salary ~ cgpa + backlogs + college_tier + country + university_ranking_band +
    internship_count + aptitude_score + communication_score + specialization +
    industry + internship_quality_score,
  data = placed_data
)

selected_model <- stepAIC(full_model, direction = "both", trace = FALSE)

model_comparison <- tibble(
  model = c("full_model", "selected_model"),
  aic = c(AIC(full_model), AIC(selected_model)),
  r_squared = c(summary(full_model)$r.squared, summary(selected_model)$r.squared),
  adjusted_r_squared = c(summary(full_model)$adj.r.squared,
    summary(selected_model)$adj.r.squared)
)

write.csv(tidy(full_model), "output/tables/full_model_summary.csv", row.names = FALSE)
write.csv(model_comparison, "output/tables/model_comparison.csv", row.names = FALSE)
write.csv(tidy(selected_model), "output/tables/final_model_coefficients.csv", row.names = FALSE)
write.csv(as.data.frame(vif(selected_model)), "output/tables/vif_selected_model.csv")
```

Code A3. Interaction check, residual diagnostics, and transformation check

```
library(dplyr)
library(broom)
library(ggplot2)
library(scales)
library(MASS)

base_model <- lm(
  salary ~ cgpa + aptitude_score + college_tier + country +
    university_ranking_band + specialization + industry,
  data = placed_data
)

interaction_model <- lm(
  salary ~ cgpa + aptitude_score + college_tier * country +
    university_ranking_band + specialization + industry,
  data = placed_data
)

ols_model_comparison <- tibble(
  model = c("base_ols", "interaction_ols"),
  aic = c(AIC(base_model), AIC(interaction_model)),
  r_squared = c(summary(base_model)$r.squared, summary(interaction_model)$r.squared),
  adjusted_r_squared = c(summary(base_model)$adj.r.squared,
    summary(interaction_model)$adj.r.squared)
)

interaction_anova <- anova(base_model, interaction_model) %>% tidy()

boxcox_result <- MASS::boxcox(interaction_model, plotit = FALSE)
best_lambda <- boxcox_result$x[which.max(boxcox_result$y)]

write_csv(ols_model_comparison, "output/tables/ols_model_comparison.csv")
```



```
write_csv(interaction_anova, "output/tables/ols_interaction_anova.csv")
write_csv(tibble(best_lambda = best_lambda), "output/tables/boxcox_summary.csv")
```

Code A4. Final model used for interpretation

```
library(tidyverse)
library(broom)
library(scales)

placed_data <- read_csv("data/placed_salary_data.csv", show_col_types = FALSE) %>%
  mutate(
    college_tier = factor(college_tier, levels = c("Tier 1", "Tier 2", "Tier 3")),
    country = factor(country, levels = c("Canada", "Germany", "India", "UK", "USA")),
    university_ranking_band = factor(university_ranking_band,
                                     levels = c("100-300", "300+", "Top 100")),
    specialization = factor(specialization,
                           levels = c("AI/ML", "Cloud", "Core CS", "Cybersecurity", "Data Science")),
    industry = as.factor(industry)
  )

final_model <- lm(
  salary ~ cgpa + aptitude_score + college_tier * country +
    country * specialization + university_ranking_band + industry,
  data = placed_data
)

final_model_fit_summary <- tibble(
  model = "final_ols_country_tier_country_specialization",
  n = nobs(final_model),
  residual_df = df.residual(final_model),
  r_squared = summary(final_model)$r.squared,
  adjusted_r_squared = summary(final_model)$adj.r.squared,
  residual_standard_error = sigma(final_model),
  aic = AIC(final_model),
  bic = BIC(final_model)
)

final_model_coefficients <- tidy(final_model, conf.int = TRUE)

write_csv(final_model_fit_summary, "output/tables/final_model_fit_summary.csv")
write_csv(final_model_coefficients, "output/tables/final_model_coefficients_clean.csv")
```

Code A5. Predicted salary tables for interpreting interactions

```
baseline_values <- tibble(
  cgpa = median(placed_data$cgpa, na.rm = TRUE),
  aptitude_score = median(placed_data$aptitude_score, na.rm = TRUE),
  university_ranking_band = names(sort(table(placed_data$university_ranking_band), decreasing = TRUE))[1],
  industry = names(sort(table(placed_data$industry), decreasing = TRUE))[1]
)

country_tier_predictions <- expand_grid(
  country = levels(placed_data$country),
  college_tier = levels(placed_data$college_tier),
  specialization = names(sort(table(placed_data$specialization), decreasing = TRUE))[1]
) %>%
  mutate(
    cgpa = baseline_values$cgpa,
    aptitude_score = baseline_values$aptitude_score,
    university_ranking_band = baseline_values$university_ranking_band,
    industry = baseline_values$industry,
    predicted_salary = predict(final_model, newdata = .),
    predicted_salary_display = dollar(predicted_salary, accuracy = 1)
  )

country_specialization_predictions <- expand_grid(
  country = levels(placed_data$country),
  specialization = levels(placed_data$specialization),
  college_tier = "Tier 1"
) %>%
```

```

mutate(
  cgpa = baseline_values$cgpa,
  aptitude_score = baseline_values$aptitude_score,
  university_ranking_band = baseline_values$university_ranking_band,
  industry = baseline_values$industry,
  predicted_salary = predict(final_model, newdata = .),
  predicted_salary_display = dollar(predicted_salary, accuracy = 1)
)

write_csv(country_tier_predictions,
  "output/tables/final_model_predicted_salary_by_country_tier.csv")
write_csv(country_specialization_predictions,
  "output/tables/final_model_predicted_salary_by_country_specialization.csv")

```

Code A6. Final-model residual diagnostics figure

```

dir.create("output/figures", recursive = TRUE, showWarnings = FALSE)

png("output/figures/final_model_residual_diagnostics.png",
  width = 1200, height = 1000, res = 120)
par(mfrow = c(2, 2))
plot(final_model)
dev.off()

resid_final_data <- augment(final_model) %>%
  mutate(studentized_residual = rstudent(final_model),
    leverage = hatvalues(final_model),
    cooks_distance = cooks.distance(final_model))

write_csv(resid_final_data, "output/tables/final_model_residual_diagnostics.csv")

```